A QUASI-COMPACT MORPHISM OF ALGEBRAIC SPACES INDUCES A QUASI-COMPACT MAP OF TOPOLOGICAL SPACES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is a topological translation of the definition of quasi-compactness of a morphism; the description of the points of the fibre product is what identifies the preimage.

Editorial note. The five references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Every open subset of |Y| is the underlying set of an open subspace. Properties of Spaces, Lemma spaces-properties-lemma-open-subspaces, used at proof lines 1274-1276. Turns the quasi-compact open V into an open subspace Y'.
- Quasi-compactness of an algebraic space tested on the underlying topological space. Properties of Spaces, Lemma spaces-properties-lemma-quasi-compact-space, used at proof lines 1277-1279 and 1285-1287. Converts topological quasi-compactness into quasi-compactness of Y' and back for |X'|.
- The definition of quasi-compactness for a morphism of algebraic spaces. Definition definition-quasi-compact, used at proof lines 1279-1280. Makes X' = Y' x_Y X quasi-compact.
- Base change of open immersions and the description of points of a fibre product. Spaces, Lemma spaces-lemma-base-change-immersions and Properties of Spaces, Lemma spaces-properties-lemma-points-cartesian, used at proof lines 1281-1284. Identifies X' as an open subspace of X and |X'| with the preimage of V.

Source reference. Original definition 03HE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4563–4563. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02YW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3891–3891. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 03BZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4366–4366. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03H4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4553–4553. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03E4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4443–4443. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. If $f : X \to Y$ is a quasi-compact morphism of algebraic spaces over S, then the underlying map $|f| : |X| \to |Y|$ of topological space is quasi-compact.*

Proof. Let $V \subset |Y|$ be quasi-compact open. By Properties of Spaces, Lemma 03BZ there is an open subspace $Y' \subset Y$ with $V = |Y'|$. Then Y' is a quasi-compact algebraic space by Properties of Spaces, Lemma 03E4 and hence $X' = Y' \times_Y X$ is a quasi-compact algebraic space by Definition 03HE. On the other hand, $X' \subset X$ is an open subspace (Spaces, Lemma 02YW) and $|X'| = |f|^{-1}(|X'|) = |f|^{-1}(V)$ by Properties of Spaces, Lemma 03H4. We conclude using Properties of Spaces, Lemma 03E4 again that $|X'|$ is a quasi-compact open of $|X|$ as desired. □